

GSC101 MID Term

The anomalous behavior of water can be seen at temperature.....

0 K

4 C

32 F

100 C

Equation of motion can only use to find the solution of problems for.....

Motion along the circular

pathMotion in straight line

Motion along curved path

All of above

For the sequence of natural if n natural numbers = summation $(n)=21$ then value of n is

-7

-6

6

7

If the sum of first n natural is $n(n+1)/2$, then their average will be.....

$n/2$

$(n-1)/2$

$(n+1)/$

2

$$N(n+1)/2$$

If an incident ray passes through the center of curvature of a spherical mirror, the reflected ray will: Pass through the focus

Pass-through the center of

curvature Through pass the pole

A car travelling at 90KM/hr on highway overtaking a bus with 120Km/hr speed. What is relative velocity of car with respect to passenger sitting in bus ?

30km/hr

210

meters

90km/hr

120km/hr

Between the melting point of ice and the boiling point of water

there are 273 Kelvin

110 degrees Fahrenheit

373 degrees centigrade

180 degrees Fahrenheit

The change in length of an object when it undergoes a temperature change is related to Its original length

Its mass

The magnitude of the temperature

change (i) only

(ii) & (iii) 0

nly

(i) & (iii) on

ly

(i), (ii) & (iii)

The object emits the light and become source of light are called.....object
Transparent

Opaque

Luminous

Translucent

t

If we apply the force on an object but the object does not move ,we can say thatApplied force is equal to force of friction

Applied force is less than force of

friction Applied force is greater than

force of friction Applied force is equal to

force of gravity

Which of the following process does not involve any chemical reaction?
Burning of paper when heated

Melting of candle wax when

heated Ripping of banana

Digestion of food in our stomach

A stone of mass 2 kg dropped from a tower fall to the ground in 3 second .A second mass of 3 kgDropped from the same height reaches the ground in:

2 seconds

3 seconds

4.5

seconds6

seconds

Liquids and gases expand on heating ,produce :

Convection currents in fluids due to changing

masses Convection currents in fluid due to

changing densitiesHeat transfer by conduction



Convection currents in fluid due to constant temperatures

The amount of heat required to raise the temperature of 1kg mass of that substance through 1K is called its.....

Specific heat

Specific

energy Latent

heat Latent

energy

Slowing down in opposite direction to the direction of the velocity is known as Uniform acceleration

Non-uniform

acceleration Positive

acceleration Negative

acceleration

Same force F is applied on two objects X and Y but the acceleration caused by this applied F are not same. The object X has acceleration three times that of object Y because -----

X is times more massive

than Y X has one-third the

mass of Y

X has different shape so easily

moved Y has more value of friction

If the object is placed -----, then the image formed by a concave mirror is virtual, erect and Magnified;

Between the pole of the mirror and the

focus Beyond the center of curvature

At the centre of



curvatureAt the focus

For a force F acting at an angle? Above the horizontal the component along the X
And the Y axis are equal . The angle?in degrees between the force and the
horizontal is ;0

30

45

60

if we push the ground in the backwards direction and the ground exerts a force in the
forwardforward direction enabling us to walk .This process is described by which law
law of Inertia

law of action and reaction

law of conservation of

momentumthe force and

acceleration law

when mercury level rises in the clinical thermometer, the mercury in capillary
tube doesnot contract

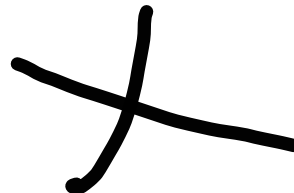
mercury takes a long time to contract

the amount of mercury use is very

small

the capillary tube has a small constriction near the

bulbthe capillary tube is very narrow



A motion where a body has the exact same motion around a fixed point is an example of
Random motion

Translator

motion Vibratory

motion Circular

motion

Which of the following quantum number gives the shell number where electron

occupy?L



M

S

N

Which of following has positive charge? Atom

Electro

n

Proton

Neutro

n

Isotopes are sister atoms of same elements with similar chemical properties But different -----

Atomic structure

Atomic volum

Atomic

numbers

Atomic weights

A person suffering from long -sighted cannot see objects nearer to his eye than 50 Cm.

He wants to read a book 25cm away; he should use spectacle lenses whose power in diopters is

.....

-6

-4

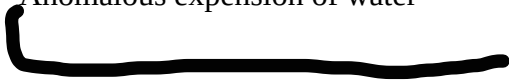
-2

In cold areas during winters pipes which are not properly insulated may burst due to: Real expansion of water

Apparent expansion of

waterExpansion of pipes

Anomalous expansion of water



According to the sign convention ,the distance of objectfrom

mirror/lens: Is always positive

Is always negative

May be positive or

negative Is equal to object

height

-----is combined with iron to make

steel Sulphur

Carbon

Iodine

Magnesium

m

Force is defined as the rate of change

of : Velocity

Acceleration

n

Momentum

Mass

Amino acids are organic compounds which contain--.

Oxygen

n

Sulphur

r

Nitrogen

n

Calcium

m

If the velocity of an object is decreasing ,acceleration has to be

Positive

Negativ

e

Unifor

m

Variable

The energy stored in compressed spring is known as -----

Kinetic energy

Chemical

energy Sound

energy

Elastic potential energy

The SI unit of energy is :

Newto

nWatt

Pascal

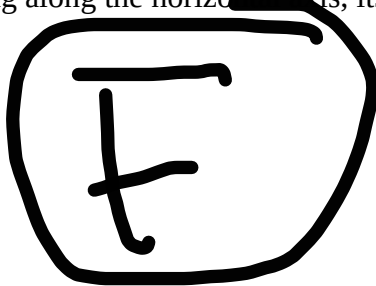
Joule

For a force F acting along the horizontal axis, its horizontal component equals: 0

$\frac{1}{2}$

F

$2F$



The most abundant metal in earth crust is -----

Calcium

Sodium

Iron

Aluminum

m

Which of the following metals are

corroded ? Cu

Fe



A

g

Ni

Which of the following is the fastest

speed 30,000 km per day

300 km per day

300 meter per second

300 millimeters per microsecond

The first step in the scientific method or process is : Predication

Observation

Hypothesis

Experimentatio

n

A person sitting inside a moving train is in a state of with respect to the other Passengers

Uniform speed

Variable speed

Uniform

acceleration

Rest

Rest

A motion of a turbine is an example of :

Random motion

Translatory motion

Vibratory motion

Circular motion

A ----- mirror is used by dentists to focus light on the tooth of a patient

Cylindric

alPlane

Convex

Concave

Valencies of noble gases in general are -----

Three

One

Two

Zero

If focal length of a lens is -50cm, the power of lens will be -----

+50 D

-2D

-50 D

+2 D

Optical fiber works on :

Principle of total internal
reflectionReflection

Refraction

Polarization

n

Efficiency is defined as:

Useful output energy / input energy

Input energy / total output energy

Total output energy / input energy

Input energy / useful output

energy

The rate of change of velocity is

known as Uniform velocity

Variable

velocity

Average

velocity

Acceleration

A motion of a child on a swing is an example of :

Random motion

Translator

motion Vibratory

motion Circular

motion

What three things can light rays travel

through? Solids, liquids, gases

Air, water, outer space

Humans, plants

, animals, Liquid, air

, plants

When two bodies are moving relative to one another the motion between them is

Translato

r Circular

Rotationa

lRelative

The short distance between two points is also known as

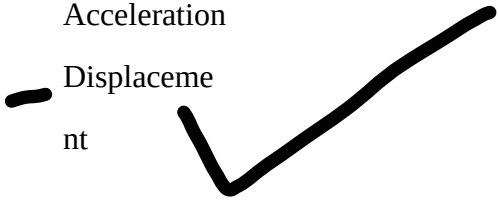
the :Distance

Speed

Acceleration

Displacement

nt



Which of the following contain -CHO functional

group ? Alcohols

Aldehyde

sKetones

Esters

Greater the value of refractive index of a medium ,the bending of light will be -----

Greater

Smaller

Zero

Negative

e

The branch of physics which deals with the study of motion of bodies without the influence of force is known as -----

Mechanics

Kinematics

Dynamics

Modern

physics

Due to which property metals are used in making of musical instrument like sitar and violin ?

Conductivity

Malleability

Ductility

Sonorousness

The objects emits the light and become source of light are called objects .

Transparent

Opaque

Luminous

Translucent

t

Solar cell technology is an example of --.

Thermal energy

Nuclear energy

Renewable energy

Non-renewable

energy

The energy of an atom in a compound --.

Remains same

Higher than

individual Lesser

than individual Can

not find out

The most abundant metal in earth crust is ---.

Calcium

Sodium

Iron

Aluminum

m

Between the melting point of ice and the boiling point of water

there are 273 kelvin

110 degrees Celsius

180 degrees Fahrenheit

373 degrees centigrade

Due to which property metals are used in making of musical instruments like sitar and violin ?

Conductivity

Malleability

Ductility

Sonorousness

5

Which line represents the SI unit of force and its

unit : Kilogram meter kg m

Gram g

Kilogram

kg Newton

N

Newton

Acceleration of a car moving steadily from 2m/s to 10 m/s in 50 s is --.

0.24m/s²

0.4m/s²

0.61m/s²

1.6m/s²

Which of the following halogens element has highest electron affinity ? Fluorine

Chlorine

Bromine

Iodine

Till now , how many elements are known ? 92

118

112

124

The SI unit of heat

is :Kelvin

Joule

Thermomete

rWatt

Which of the following is a derived unit:

Kilogra

m

Kelvin

Ampere

Newton

A kilowatt hour is a unit

ofPower

Force

Energ

y

Time

Sum of first 2000 term of the sequence : $\{-1, 1, -1, 1, \dots\}$ is -----

-1

1

0

Undefined

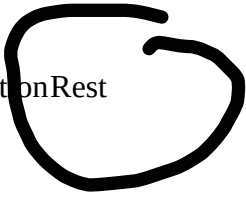
A person sitting inside a moving train is in a state ofwith respect to other

passengersUniform speed

Variable speed

Uniform

acceletationRest



Solar cell technology is an example of --.

Thermal energy

Nuclear energy

Renewable energy

Non-renewable

energy

The branch of physics which deals with the study of motion of bodies without the influence of Force is known as --.

Mechanics

s

Kinematics

cs

Dynamics

Morden physics

There is no deviation in the path of ray of light when it passes through ____ of a lens

Optical

centerFocus

Center of

curvature Pole

Work done depends upon (i) force (ii) distance moved in direction of force applied and (iii) angle between direction of force and displacement .which line is always correct if workDone is not zero :

(i) is not zero

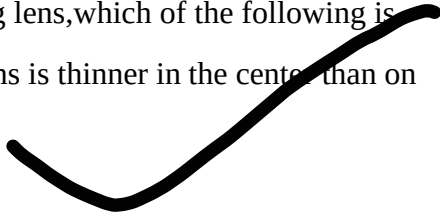
(i) and (ii) are not

zero (i) and (iii) are

not zero

(ii) and (iii) are not zero

For diverging lens, which of the following is true ? The lens is thinner in the center than on the edges.



The lens is thicker in the center than on the edges. Light must enter it parallel to the principle axis .

The lens must be flat on one side ,

Which of the following statement /s is /are correct :

Optical fibers can bend light using the principle of laws

Optical fibers bend the light beam using the principle of total internal reflection

Optical fibers cannot bend the light

Optical fibers can bend light using the principle of laws of reflection

Sum of the geometric series : $1 + (1/2) + (1/4) + (1/8) + \dots$ is

2

$3/2$

1

$5/2$

When a horse pulls a wagon the horse moves forward due to the force : Of the horse on the wagon

Of the wagon on the

horse Of the horse on the

ground

Of the ground on the horse

In Prokaryotic cells the outer most layer is known as ____.

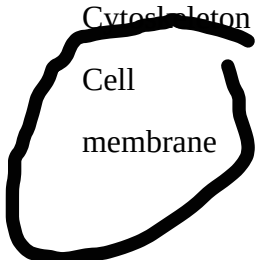
Cytoplasm

Cell wall

Cytoskeleton

Cell

membrane



What important activity takes place in the lungs

Oxygen is exchanged for carbon dioxide
The trachea is exchanged for the larynx
Food is digested
Liquid waste is filtered from the blood

The main function of cell membrane is_____.

Helping in movement of organelles
Providing space for metabolic reaction

Transporting of materials in and out from cell
Giving protection strength and support

What is the main job of the red corpuscles in the blood?

To transport oxygen to the body's cells and carry away carbon dioxide from the cells
To clot blood

To fight disease

To transport carbon dioxide to the body's cells and carry away oxygen from the cells

In anaphase I the chromosomes contain one pair of_____.

Daughter chromatids

Sister nucleosomes

Sister chromatids

Daughter

nucleosomes

Mitochondria are called powerhouse of a cell because_____.

They make ATP (Adenosine Tri Phosphate) for

cell
They have a double membrane

They present in cytoplasm in large

number
They are self replicating

organelle

During Anaphase of mitosis the chromosomes are_____.

Moving towards the poles

Reached at the poles

Arranging themselves at the

equatorReplicating themselves

Each distinct form of life or type of organism is called _____

Species

Cell

System

Domain

n

What is the respiratory

system?The body's system of

nerves The body's breathing

system

The body's blood-transporting

systemThe body's food-processing

system

Products excreted by the skin are excess water, urea, uric acid
and salt
hairs

carbon

dioxide oil

Another bladder in the body apart from the urinary bladder is
the: Bile duct
Prostate bladder

Gal bladder

Kidney
artery

What is the respiratory system?

The body's food-processing
system

The body's system of nerves

The body blood-transporting system

The body's breathing system

Animals such as bears that consume both plants and other animals
Producers

Herbivores

Omnivore

s

Carnivore

s

Flowers have male and female parts called

.....and.....Pollination carpel

Stamen

pollinationCarpel

stamen

Stamen carpel

How many chromosomes does a normal, mature, human Sperm cell contain ? 21

22

23

46

The smallest blood vessels are_____.

Arteries

Veins

Capillaries

es

Arteriole

s

What important activity takes place in the lungs?

Oxygen is exchanged for carbon

dioxide Liquid waste is filtered from

the blood The trachea is exchanged

for the larynx Food is digested

The entire planet and all of its living and nonliving parts

Biosphere

Community

Ecosystem

Population

Population

The removal of the metabolic wastes of an

organism is: Respiration

Excretion

Defecation

Renal

artery

A group of organisms of the same species that interbreed and live in the same place

Population

Community

Ecosystem

Organism

Species

Organisms that use energy from The sun to make their own

food are Carnivores

Consumers

Producers

Omnivore

s

Air can enter the body and travel to the lungs
Through the esophagus and gullet
Through the windpipe and the pores

Through the mouth and the nose

Through the nose and the nervous system

What is the purpose of the little hairs inside the nose?
To tickle the nose and cause sneezes

To keep dust out of the

lung They serve no

purpose

To fight disease

Animals such as foxes that consume other animals for their food and energy

Carnivore

s

Producers

Herbivore

s

Omnivore

s

Which one in blood help in exchange of oxygen against

Contraction? White blood cells

Plasm

a Salts

Hemoglobin

What is the organ that pumps blood all throughout the human body?
The blood vessels and capillaries

The

kidneys

The lungs

The heart

The removal of the metabolic wastes of an organism is: Respiration

Excretion

Defecation

Renal

artery

Products excreted by the skin are excess water, urea, uric acid and

salt

carbon

dioxide oil

Hair

Seed is formed from the.....

Carpel

Pollination

n

Zygote

Stamen

n

Why is blood that flows from the lungs to the heart bright red rather than dark red? The lungs add a pigment (dye) to blood as it flows through them

Gastric juices produce the red color of the blood. Carbon dioxide makes it red

Oxygen makes it red

Left ventricle pushes blood towards

.....Heart

Tricuspid valve

Aort

a

Lung

s

Right ventricle pushes blood in the pulmonary artery

Towards....Bicuspid valve

Tricuspid

valveHearts

Lungs

The tiny air sacs in lungs are called ____

Bronchiole

sRavioli

Alveo

li

Bronx

A collection of interacting

populationsSpecies

Population

Communit

y

Organism

Another bladder in the body apart from the urinary bladder Is

the:Bile duct

Prostate

bladderGall

bladder

Kidney artery

What is the main job of the red corpuscles in the blood? To clot blood

To fight disease

To transport carbon dioxide to the body's cells and carry away oxygen from the cells

To transport oxygen to the body's cells and carry away Carbon dioxide from the cells

The organ that produces urea and acid as a by-product of the breakdown of proteins is the :Stomach

Liver

Small

intestine

Kidney

The highest blood pressure is found in the _____

Venue

sAorta

Arterie

sVeins

One degree of Celsius is equal to?

One degree of Fahrenheit scale

1.8 degree of Fahrenheit scale

~~3.2 degree of Fahrenheit scale~~

2.12 degree of Fahrenheit scale

The aluminum metal is used for making cooking utensils because metals are _____.

Good conductors

~~malleable~~

Ductile

Solid at room temperature

A hydroelectric power plant converts the gravitational potential energy of water to electric energy

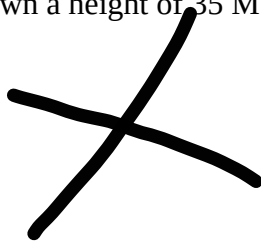
.if 150,000kg of water flowing down a height of 35 MW. The efficiency of the system is

50 to 55%

56 to 62%

75 to 85%

93 to 97%



A mass is projected vertically downwards. Just at the point of release with the ground its total energy is :

- 0% gravitational potential and 100% kinetic energy
- 25% gravitational potential and 75% kinetic energy
- 50% gravitational potential and 50% kinetic energy
- 100% gravitational potential and 0% kinetic energy

Mercury is chosen as a thermometric substance because: i. over a wide range of temperature its expansion is uniform ii. it does not stick to thermometer glass iii. It is opaque to light

- (i) Only
- (ii) & (iii) only
- (i) & (iii) only
- (i), (ii) & (iii)

The gas law giving the relationship between 'V' and 'P' of a gas is

- Dalton's law
- Charles's law
- Graham's law
- Boyle's law

The graph between V and $1/p$ at constant temperature is:

- A straight line
- Parabola
- A curve of any shape
- Hyperbola

Which of the following halogens element has highest electron affinity?

- Fluorine
- Chlorine
- Bromine
- Iodine

Metals are usually _____

- Hard
- Hard and ductile
- Ductile and malleable
- Hard, ductile and malleable

Force is defined as the rate of change of:

- Velocity
- Acceleration
- Momentum
- Mass

_____ is used as a tonic medicine and in the manufacturing of ink and pigments

- Ferrous sulphate
- Ferrous fluoride
- Ferric oxide
- Ferrosoferric oxide

A motor pumps 4,000kg of water from a ground tank to an overhead tank 50 m up in one hour.

The power of the motor is:

55.5W

250 W

555W

33,333W

Two rectangular strips of copper and steel are wrapped up to form a bi-metal strip. On heating the bi-metal strip will:

Contract

Remain straight

Bend with steel on the convex side

Bend with steel on the concave side

When two equal and opposite vectors are subtracted from one another the resultant is

Zero

Half of the original vector

The original vector

Twice the original vector

Isotopes are sister atoms of same elements with similar chemical properties but different

Atomic structure

Atomic volume

Atomic numbers

Atomic weights

Another statement of newton second law of motion is ‘rate of change of momentum is equal to’

Acceleration

Mass

Force

Velocity

Energy possessed by an object due to its motion is termed as

Potential energy

Motion energy

Thermal energy

Kinetic energy

The temperature of sand raised more quickly as compared to water as specific heat of sand is

Less than water

Equal to water

Greater than water

Neutral

According to the sign convention, the distance of object.....from mirror/lens:

Is always positive

Is always negative

May be positive or negative

Is equal to object height

A car travels at 10km/hr for 2 hours and then at 13km/hr for 1 hour. What is the value of average speed in km/hr

- 8km/hr
- 11km/hr
- 15km/hr
- 17km/hr

For a force F acting along the horizontal axis, its vertical component equals

- 0
- $\frac{1}{2}F$
- F
- $2F$

A battery of car has stored Energy, which it converts to electric energy when required.

- Chemical energy
- Potential energy
- Thermal energy
- Kinetic energy

Matter can exist in four states (i) gas, (ii) plasma (iii) solid and (iv) liquid. For a particular element normally the states arranged in order of rising temperature (lowest to highest) are:

- (i),(ii),(iii) and (iv)
- (iv),(iii),(ii) and (i)
- (ii),(i),(iv) and (iii)
- (iii),(iv),(i) and (ii)

The reading on the Fahrenheit scale will be double the reading on the centigrade scale when the temperature on the centigrade scale is?

- 160
- 280
- 460
- 360

What is the name of chemical bond between two atoms where both bonding electrons donated by a single atom?

- Ionic bond
- Covalent bond
- Coordinate covalent bond
- Hydrogen bond

The value of the coefficient of thermal expansion of a solid depends on the value of:

- The unit of length as well as of temperature
- The unit of length and not of temperature
- The unit of temperature and not of length
- Neither the unit of length nor of temperature

An image formed by a convex mirror is always:

- Virtual, erect and diminished
- Virtual, real and magnified
- Real, erect and magnified

Another statement of newton second law of motion is "rate of change of momentum is equal to _____",

Acceleration

Mass

Force

Velocity

Which of the following is a true statement?

The power of a lens is always positive.

The power of lens is always negative.

The power of a convex lens is positive.

The power of a concave lens is positive.

The composition (by mass) of Fe in stainless steel is _____ percent

80.6%

76%

96%

98%

Which of the following is not an electromagnetic wave?

X-rays

Ultraviolet

Sound

Visible light

Which of the following group contain alkaline earth metals

Group 1

Group 2

Group 3

Group 7

Work done depends upon (i) force (ii) distance moved (iii) angle between direction of force and displacement. Which line is correct if work done is not zero: force angle displacement?

Zero zero degree zero

Zero 90 degrees zero

Greater than zero degrees greater than

Greater than 90 degrees greater than zero

In using the equations of motion all quantities having direction same as the initial velocity are taken as

Positive

Negative

Zero

Uniform

Sum of all type of energy that an object can possess is called _____.

Temperature

Power

Heat

Density

If the thickness of human eye lens increases, the image will formed _____.

Behind the retina
In front of retina
On retina
In between lens and retina

Due to anomalous expansion liquid-water has a volume increase as it cools below:

10
4
-4
-10

Which line correctly represents mass and weight

Mass weight
Is a vector quantity unit is kilogram
Is a scalar quantity unit is newton
Unit is newton is a vector quantity
Unit is kilogram is scalar quantity

A motorcycle was moving with uniform acceleration of 6m/s^2 . in 10 s it will cover_____.

60 meters
100 meters
300 meters
450 meters

When ray of light enter from denser medium to rare medium it bends_____.

Away from normal
Towards normal
Perpendicular to normal
Parallel to normal

Inverse of focal length is known as the _____ of a lens:

Focus
Power
Power of accommodation
Far point

Distance covered in unit time is:

Displacement
Speed
Velocity
Acceleration

Science does not deal with:

Observation
Experimental
Hypothetical assumptions
Subjective evidence

The M.P caustic is _____degree centigrade

100
801

95

318

A car with uniform speed can have the acceleration only when the motion of car is in_____.

Straight line

Circular path

Irregular path

Parabolic path

The rate of change displacement is known as:

Speed

Velocity

Acceleration

Retardation

If the body covers equal distances in equal intervals of time, irrespective of the interval, that body is said to be:

Accelerating

Moving with uniform velocity

Moving with uniform speed

Decelerating

The amount of work done by a Force of 5 N pulling a mass of 5 kg, 5m along a rough floor in the direction of the applied force is

5 J

25 J

50 J

125 J

When water is heated from 0 to 20 C its volume

First decreases and then increase

Goes on decreasing

Remains constant up to 4 C and then increase

A battery of car has storedenergy, which it converts to electric energy when required.

Chemical energy

Potential energy

Thermal energy

Kinetic energy

White light separates into different colors after passing through a -----:

Mirror

Camera

Prism

Hammer

The atom, which provides the electron pair, is called_____

Gainer

Acceptor

Donor

Receiver

The refractive index of a denser medium with respect to a rare medium is-----:

1

Greater than 1

Smaller than 1

Negative

Which of the metal exists in liquid state at room temperature?

Fe (Iron)

Cu (copper)

Hg (mercury)

Al (aluminum)

The shielding effect in Na is

Greater than that of Li

Greater than that of K

Equal of that K

Equal of that Li

The metal reacting readily with cold water is.

Au

Ag

Na

Mg

Find the power of a convex lens, if its focal length is 25cm:

2.5 D

4 D

0.4 D

0.25 D

White light separates into different colors after passing through a -----:

Mirror

Camera

Prism

Hammer

If a bus is moving with uniform velocity on highway, it can be concluded that_____.

It is moving with uniform acceleration

There are no forces acting on it

There are no unbalanced (net) forces acting on it

It has large value of inertia

The example of giant molecule is_____

Sic

Diamond

AlN

AgCl

The rate of reaction of organic compounds is very slow due to _____.

Ionic bonding

Covalent bonding
Coordinate covalent bonding
Metallic bonding

Water has an expansion which is considered as an anomalous behavior at which temperature range:

0 K to 10 K
273 K to 283 K
373 K to 383 K
-10 K to 0 K

One degree of Celsius is equal to?

One degree of Fahrenheit scale
1.8 degrees of Fahrenheit scale
3.2 degrees of Fahrenheit scale
2.12 degree of Fahrenheit scale

If two quantities are directly proportional to each other then the graph between them is a:

Circle
Arc
Straight line
None of the above

The refractive index of a denser medium with respect to a rarer medium is-----:

1
Greater than 1
Smaller than 1
Negative

Image that can be obtained on a screen:

Virtual
Real
Diverging
Converging

The purest form of iron produced when impurities are removed is known

Pig Iron
Cast Iron
Wrought Iron
Steel

Working principle of geyser based on which of the following?

Conduction
Convection
Radiation
Charles law

The temperature of sand raised more quickly as compared to water as specific heat of sand is_____.

Equal to water
Greater than water
Less than water

Neutral

If a body does not change its position with respect to the observer than it is said to be in a state of:

Rest

Motion

Acceleration

Retardation

Two rectangular strips of copper and steel are wrapped up to form a bi-metal strip. On heating the bi-metal strip will:

Contract

Remain straight

Bend with steel on the convex side

Bend with steel on the concave

Matter can exist in four states (i) gas (ii) plasma, (iii) solid and (iv) liquid. For a particular element normally the states arranged in order of falling temperature (highest to lowest) are:

(i),(ii),(iii)and (iv)

(iv),(iii),(ii)and (i)

(ii),(i), (iv) and (iii)

(iii),(iv),(i) and (ii)

An image formed by a convex mirror is always:

Virtual, erect and diminished

Virtual, real and magnified

Real, inverted and diminished

Real, erect and magnified

The power of a machines do a work of 100 J in 10 seconds is _____.

10 watt

100 watt

500 watt

1000 watt

A person walks 3 km north, then moves 4 km east. His displacement is

3 km East of North

4 km East of North

5 km East of North

7 km East of North

If the temperature of a gas increased by keeping the volume constant, then:

Its pressure rises

Its pressure is constant

Its pressure falls
Suddenly increases and then decreases

The cell in which ~~electrical~~ energy is converted into chemical energy is called
Electrolytic cell
Galvanic cell
Daniel
Fuel cell

Water has an expansion which is considered as an anomalous behavior at which temperature rang:
0 K to 10 k
273 k to 283 k
373 k to 383 k
-10 k to 0 k

The composition (by mass) of Fe in stainless steel is _____ percent
80.6%
67%
96%
98%

The force produced due to gravity is known as _____.
Mass
~~Tension~~
~~Weight~~
Friction

The SI unit of temperature is:
Kelvin
Celsius
Fahrenheit
Centigrade

If two quantities are inversely proportional to each other than on the graph when one quantity increases the other
Increases
Decreases
Remains constant
Increases and then decreases

In the optical fiber the refractive index of core in comparison to that of cladding is:
Smaller
Greater
Equal
Minor

When a person uses a convex lens a simple magnifying glass, the object must be placed at a distance:
Less than one focal length
More than one focal length
Less than twice the focal length

More than twice the focal length

Distance covered in unit time is:

Displacement

Speed

Velocity

Acceleration

Molecule has a polar covalent bond?

NaCl

HCl

H₂

None of these

Total internal reflection occurs when:

Light passes from air into water

Light refracts as it exits glass into air

Light passing through glass is reflected inside the glass

The angle of incidence is less than critical angle

Simple thermostat works on:

Condensation

Specific heat

The second law of thermodynamics

Thermal expansion

Example of derived unit is _____.

Current

Speed

Temperature

Quantity of a substance

The rate of change of velocity is known as

Uniform velocity

Variable velocity

Average velocity

Acceleration

A _____ mirror is used by dentists to focus light on the tooth of a patient

Cylindrical

Plane

Convex

Concave

When light travels from glass to air, the speed of light will _____,

Decreases

Increases

Not change

Depend on angle of incidence

Inverse of focal length is known as the _____ of a lens:

Focus

Power

Power of accommodation

Far point

Two rectangular strips of copper and steel are wrapped up to form a bi-metal strip. On heating, the metal strip will:

Contract

Remain straight

Bend with steel on the convex side

Bend with steel on the concave

A straight line graph proportional between displacement and time indicates

Average speed

Uniform speed

Average velocity

Uniform velocity

Which of the metal exists in liquid state at room temperature?

Fe (Iron)

Cu (copper)

Hg (mercury)

~~Al (aluminum)~~

Temperature is a property which determines:

How much heat a body contains

Whether a body will feel hot or cold to touch

In which direction heat will flow between two systems

How much total absolute energy a body has

A man started his walk from main gate of park and after completing two rounds of track of 100 meters in 10 minutes he came back to the main gate. What is value of his total displacement?

Zero

10 meters

100 meters

200 meters

Working principle of geyser based on which of the following?

Conduction

Convection

Radiation

Charles law

Optical instrument like telescope, microscopes and optical fibers are based on which light phenomena?

Absorption of light

Different of light

Total internal reflection of light

Scattered of light

Which of the following statements/s is/are correct:

Optical fibers can bend light using the principle of laws of refraction

Optical fibers bend the light beam using the principle of total internal reflection

Optical fibers cannot bend the light

Optical fibers can bend light using the principle of laws of reflection

When two equal and opposite vectors are subtracted from one another the resultant is

Zero

Half of the original vector

The original vector

Twice the original vector

Two balls of different masses, and hence of different sizes, but of the same material, are thrown upwards at the same speed. They reach the maximum height and fall back to the ground ignoring air resistance the balls will fall to the ground

Simultaneously

Heavy ball first

Lighter ball first

Larger ball fast

A vector is multiplied by four .the resultant will beTime the original vector

2

3

4

5

To get a real and inverted image of the same size using a convex lens where should an object be placed?

Between 0 and F

At F

At 2F

At infinity

The atom, which provides the electron pair, is called _____

Gainer

Acceptor

Donor

Receiver

If two quantities are inversely proportional to each other than on the graph when one quantity increases the other

Increases

Decreases constant

Increases and then decreases

A person sitting in his office reading a lengthy report comprising 800 pages is according to physics definition of work is:

Doing work

Not doing work

Gaining energy

Losing energy

The gas law giving the relationship between V and P of gas is

Dalton's law
Charles's law
Graham's law
Boyle's law

Which line correctly identifies the steam point on the scales of temperature: kelvin/K Celsius/
Fahrenheit/

273 0 32

273 100 212

373 0 32

373 100 212

Which line correctly identifies average speed?

Distance covered in unit time

Displacement in unit time

Distance covered in constant time intervals

Total distance covered divided by time taken

Sum of all types of energy that an object can possess is called

Temperature

Power

Heat

Density

If the pressure of a given is held constant its density is inversely proportional to its absolute temperature .we can refer it as another statement of?

Boyle's law

Charles law

Ideal gas law

Avogadro's' law

The ring shape unsaturated hydrocarbon are known as

Aliphatic compounds

Aromatic compounds

Acyclic compounds

Isomers compounds

Find the power of convex lens, if it's focal length of is 25cm

2.5 D

4 D

0.4 D

0.25 D

A man started his walk from main gate Of Park and after completing two rounds of track of 100 meters in 10 minutes he came back to the main gate .what is value of his total displacement ?

Zero

10 meters

100 meters

200 meters

Tension in string due to gravity at always equal to _____of an object.

Mass
Weight

~~Momentum~~

Density

When ray of light enter from denser medium to rare medium it bend _____.

Away from normal

Towards normal

Perpendicular to normal

Parallel to normal

The rate of change of displacement is known as:

Speed

Velocity

~~Acceleration~~

Retardation

Metals are good conductor of heat and electricity because metals _____.

Gain electrons easily

Lose electrons easily

Have higher electronegativity

~~Are good oxidizing agent~~

When experimental results are in accordance with theoretical prediction this become:

A law

A theory

A hypothesis

An assumption

A fundamental quantity e.g. a length of one meter, has:

No error as it is by definition a perfect value

No error as it is an experimentally verified quantity so perfect

Error as it is an experimentally verified quantity so imperfect

Error as it is a theoretical value and so cannot be perfect

In H₂O every bond has ____ ionic and ____ covalent character (percent)

70% 30%

66% 33%

50% 50%

40% 60%

The M.P of caustic soda is _____ degree centigrade

100

801

95

318

Concave mirror formed an inverted image if distance between mirror and object is _____.

Less than its focal length

Exactly at its focal length

More than its focal length

None of above

If under the action of a force on an accelerating body, the work done is maximum the direction of force and the displacement would be _____ to each other

Opposite

Perpendicular

Identical

~~Insufficient information for an answer~~

Mass of astronaut is $m=50\text{kg}$. what is astronaut's mass on moon where value of $g=1.6\text{m/s}^2$?

80 kg

~~50 kg~~

100 kg

32 kg

Some of energy radiated by sun is captured by earth but the normal surface temperature of earth remains about 300K. why does not the earth's temperature rise as it captures the sun's energy?

The earth radiates an amount of energy into space equal to the amount it receives.

The energy only raises the temperature of the upper atmosphere and never reaches the surface.

The earth reflects the sun's light.

The heat is carried away from the earth by convection currents.

The direction of the change of momentum is the same as the:

Displacement

Speed

Mass

Force

Which of the following is the fastest speed

~~30,000 km per day~~

300 km per minute

300 meters per second

300 millimeters per microsecond

A battery of car has stored ----- energy, it converts to electric energy when required.

Chemical energy

Potential energy

Thermal energy

Kinetic energy

A motion of a turbine is an example of:

Random motion

Translator motion

Vibratory motion

Circular motion

Among the equation of linear motion the time independent equation(s) is (are:

(i) and (ii)

(i) Only

(ii) and (iii)

(iii) Only

In Comp

etc

If pressure of gas increases 2 times at constant temperature, the final volume of gas would be

2 v

$\frac{1}{2}$ v

3 v

$\frac{1}{4}$ v

A plane flies west at a speed of 300 km per hour for one hour. It then turns south and flies at the same speed of 300 km per hour for a further $1\frac{1}{2}$ hours. Its average speed is:

100 km per hour

200 km per hour

300 km per hour

450 km per hour

Circular motion is considered as _____ where movement is along circular path.

Translatory motion

Rotatory motion

Vibratory motion

Linear motion

In optical fiber the refractive index of core in comparison to that of cladding is:

Smaller

Greater

Equal

Minor

What three things can light rays travel through?

Solids, liquid, gases

Air, water, outer space

Humans, plants, animals

Liquid, air, plants

A derived unit is created when:

Two or more fundamental units are added or subtracted

Two or more fundamental units are multiplied or divided

Two or more fundamental units are added and divided

Two or more fundamental units are subtracted and multiplied

The elements of group 18 are called _____

Normal element

Typical elements

Noble gases

Transition elements

A straight-line graph proportional between displacement and time indicates

Average speed

Uniform speed

Average velocity

Uniform velocity

Light enters in a human eye through a transparent membrane known as _____

Retina

Cornea

Pupil

Iris

Lens used in simple microscope is:

Biconvex

Biconcave

Plano convex

Cylindrical

Which of the following groups contain alkaline earth metals?

Group 1

Group 2

Group 3

Group 7

Gradient of line of velocity time graph is representing _____

Distance

Speed

Velocity

Acceleration

Circular motion is considered as _____ where movement is along circular path.

Translatory motion

Rotatory motion

Vibratory motion

Linear motion

Water has an expansion which is considered as an anomalous behavior at which temperature range :

0 K to 10 K

273 K to 283 K

373 K to 383 K

-10 K to 0 K

The self-linking property of carbon is known as _____.

Sublimation

Hydrogenation

Carbonation

Catenation

Lakes freeze from the top down due to:

The quantity of salt in water

The anomalous behavior of the expansion of water between 0°C and 4°C

The confinement properties of the boundaries of the lake

The high specific heat of water

Which of the following quantum numbers explain the shape of sub-shells in atomic orbitals?

Principal quantum number (n)

Azimuthal quantum number (l)
Magnetic quantum number(m)
Spin quantum number (s)
Magnification (positive) greater than unity point toward:
Real image
Virtual image
Neither real nor virtual image
Distorted image

An object of mass m is moving under a constant force Causing an acceleration. if the mass of the object is doubled and the force reduced to:

0.25 a

0.5 a

2a

a

if the pressure of a given gas is held constant its density is inversely proportional to its absolute temperature. we can refer its as another statement of?

Boyle's law

Charles law

Ideal gas law

Avogadro's' law

A parson sitting inside a moving train is in a state of.....with respect of the other passengers

Uniform speed

Variable speed

Uniform acceleration

Rest

The minimum distance that needed human eyes to focus on an object is_____.

10cm

15cm

25cm

30cm

Reflection happens when light strikes and then -----a surface

Bounces off

Break

Reflect

Refract

When a name of chemical bond between two atoms where both bonding electrons by a single atom?

Ionic bond

Covalent bond

Coordinate covalent bond

Hydrogen bond

Friction can be reduced through use of ball bearings because:

Ball bearing make the surface smooth

Ball bearing convert kinetic into rolling friction

Ball bearing have no friction own
Ball bearing the make surface rough allowing better grip

The reaction in which one substance is replaced by another substance is called ____.

Substitution reaction

Addition reaction

Elimination reaction

Neutralization reaction

When two equal and opposite vectors are subtracted from one another the resultant is

Zero

Half of the original vector

The original vector

Twice the original vector

The acceleration of a moving body is always in the direction of the:

Velocity

Applied force

Reactive force

Resistive force

Which of the following is the fastest speed?

30,000km per day

300km per minute

300 meter per second

300 millimeters per microsecond

A parson walks 3km north, then moves 4km east. his displacement is

3km east of north

4km east of north

5km east of north

7 km east of north

Which of the following metals are corroded?

Cu

Fe

Ag

Ni

If a body does not change its position with respect of the observer then it is said to be in a state of:

Rest

Motion

Acceleration

Retardation

A vector is multiplied by four. The resultant will be ...time the original vector

2

3

4

5

Which of the following statement/s is /are correct:

Optical fiber can bend light using the principal of law of refraction

Optical fiber the light beam using the principal of total internal refraction

Optical fiber cannot bend the light

Optical fiber can bend light using the principal of laws of reflection

The shielding effect in Na is

Greater than that of Li

Greater than that of K

Equal of that K

Equal of that Li

Gradient of line of velocity-time graph is representing _____.

Distance

Speed

Velocity

Acceleration

When light travels from glass to air, the speed of light will _____.

Decreases

Increases

Not change

Depend on angle of incidence

If the body covers equal distance in equal intervals of time, irrespective of the duration of the interval, the body is said to be:

Accelerating

Moving with uniform velocity

Moving with uniform speed

Decelerating

The unit of power is represented by

J S

J/S

W S

W/S

The color make up white light are called -----?

Prism

Energy waves

Visible spectrum

Rainbow

The non-metal which is liquid at room temperature _____.

Mercury iron bromine

Iron

Bromine

Lead

_____ is a most reactive metal.

Zn

Au

Na

Cs

Metals are usually _____

Hard

Hard and ductile

Ductile and malleable

Hard, ductile and malleable

The naturally occurring resources of metal are known as _____.

Minerals raw materials

Mineral rocks

Mineral ores

Mineral deposit

The study of motion of objects without reference to mass and force is called:

Kinematics

Dynamics

Mechanics

General relativity

Due to which property metals are used in making of musical instruments like sitar and violin?

Conductivity

Malleability

Ductility

Sound

The principal focus point of human eyes is always _____.

Behind the retina

In front of retina

On retina

In between eye ball and retina

Till now, how many elements are known?

92

118

112

124

The energy of an atom in a compound _____.

Remains same

Higher than individual

Lesser than individual

Can not find out

Two objects X and Y are moving. If the speed of X is double that of Y and mass of X is half of Y. the ratio of their kinetic energies would be:

0.25

0.75

1.00

2.00

The mass number of carbon is C-12, it represents the_____.

Total number of protons =12

Total number of electrons =12

Total number of protons and electrons =12

Total number of protons and neutrons =12

An engineer design different machine. which is the least efficient?

Input energy useful output wastage

731J 468J 263J

450J 248J 202J

255J 153J 102J

510J 357J 153J

The refractive index of a denser medium with respect to a rarer medium is-----:

1

Greater than 1

Smaller than 1

Negative

A man started his walk from main gate of park and after completing two rounds of track of 100 meters in 10 minutes he came back to the main gate. What is value of his total displacement?

Zero

10 meters

100 meters

200 meters

Metals are good conductor of heat and electricity because metals_____.

Gain electrons easily

Lose electrons easily

Have higher electronegativity

Are good oxidizing agent

If you applied a force of 200 N on an object with mass m but the object does not move. We can say that ____.

Newton second law of motion is not valid here

Newton 1st law of motion is not valid here

The object has too much mass

There is a force of 200N in the opposite direction

The blue cooler of ocean is due to_____.

Reflection of light

Refraction of light

Absorption of light

Scattering of light

Brass (Zn +Cu) is an example of

Element

Compound

Mixture
Zinc copper

A 15gm bullet is fired from a 7.5 kg gun at 380 m/s. the momentum of the gun will be:

5700 Ns

2850 Ns

5.7 Ns

2850000 Ns

A fundamental quantity e.g. A length of one meter, has:

No error as it is by definition a perfect value

No error as it is experimentally verified quantity so

Error as it is an experimentally verified quantity so imperfect

Error as it is theoretical value and so cannot be perfect

Which type of rays produced due to ionizations of atoms or molecules of the gas in discharge tube?

Negative

Positive

Neutral

No rays produced due to ionization

When experimental result are in accordance with theoretical prediction this becomes:

A law

A theory

A hypothesis

An assumption

The gas law giving the relationship between ‘V’ and ‘P’ of gas is

Dalton’s law

Charles’s law

Graham’s law

Boyle’s law

Which line correctly identifies the steam point on the scales of temperature: kelvin/k Celsius/
C

Fahrenheit/ F

273 0 32

273 100 212

373 0 32

373 100 212

The average power of an athlete of mass 60 kg in climbing a staircase 10 m high in seconds is approximately:

60 w

300 w

600 w

900 w

The number of maximum electrons in d-sub shell are



A fundamental quantity e.g. a length of one meter, has:

No error as it is by definition a perfect value

No error as it is an experimentally verified quantity so perfect.

Error as it is an experimentally verified quantity so imperfect.

Error as it is a theoretical value and so cannot be perfect

Metals are good conductor of heat and electricity because metals_____.

Gain electrons easily

Lose electrons easily

Have higher electronegativity

Are good oxidizing agent

As you move an object away from a convex mirror, its image becomes----- and moves towards--
---;

Smaller, infinity

Smaller, focus

Enlarged, infinity

Enlarge, focus

Inverse of focal length is known as the _____ of a lens:

Focus

Power

Power of accommodation

Far point

A physical quantity which requires a unit and a number to be described fully is known as a

Fundamental quantity

Vector

Derived quantity

Scalar

Example of derived unit is _____.

Current

Speed

Temperature

Quantity of substance

Specific heat capacity is the property of the substance per unit:

Mass

Temperature

Weight

Volume

When a person uses a convex lens as a simple magnifying glass, the object must be placed at a distance:

Less than one focal length
More than one focal length
Less than twice the focal length
More than twice the focal length

Lakes to freeze from the top down due to:

The quantity of salt in water
The anomalous behavior of the expansion of water between 0 C and 4 C
The confinement properties of the boundaries of the lake
The high specific heat of water

A 1000 kg car uses 250,000 J of fuel energy to increase its speed from 10 m/s to 20m/s. the efficiency of this fuel conversion:

40%
50%
60%
70%

An engineer designs different machines. Which is the most efficient?

Input energy useful output wastage

731 J 468 J 263 J
450 J 248 J 202 J
255 J 153 J 102 J
510 J 357 J 153 J

Two balls of different masses, and hence of different size, but of the same material, are thrown upwards at fall back to the ground. Ignoring air resistance the balls will fall to the ground

Simultaneously

Heavy ball first

Lighter ball first

Larger sized ball first

A battery of car has stored.....energy, which it convert to electric energy when required.

Chemical energy

Potential energy

Thermal energy

Kinetic energy

Which line represents the SI unit of force and its unit?

Kilogram meter kg m

Kilogram kg

Newton N

.....is used as a tonic medicine and in the manufacturing of ink and pigments

Ferrous sulphate

Ferrous fluoride

Ferric oxide

Ferrosoferric oxide

The fundamental unit of mass is:

Pound

Ounce

Kilogram

Gram

A body moving in a straight line is an example of:

Random motion

Translator motion

Vibratory motion

Circular motion

The amount of work done by a force of 5N pulling a mass of 5 kg, 5m along a rough floor in the direction of the applied force is

5 J

25 J

50 J

125 J

Normally when heat is added to a body its properties i.e. energy, temperature and volume changes, which line could best represent this:

Gains rises expands

Loses rises contracts

Gains falls contracts

Loses falls expands

If the body covers equal distances in equal intervals of time, irrespective of the duration of the interval, that body is said to be:

Accelerating

Moving with uniform velocity

Moving with uniform speed

Decelerating

The amount of energy released or absorbed when electrons are added to an atom is known as _____.

Ionization energy

Electronegativity

Electron affinity

Activation energy

Circular motion is considered as _____ where movement is along a circular path.

Translator motion

Rotatory motion

Vibratory motion

Linear motion

Two thin lenses of power +6 D and -2D are placed in contact with each other. Focal length of the combination is:

+ 4 m

-4 m

+ 0.25 m

-0.25 m

The self-linking property of carbon is known as _____.

Sublimation

Hydrogenation

Carbonation
catenation

A mass is projected vertically downwards. Which line could represent the correct energy conversion K.E. gravitational PE total energy
Increases increases increases

Increases decreases does not change

Decreases decreases decreases

Decreases increases does not change